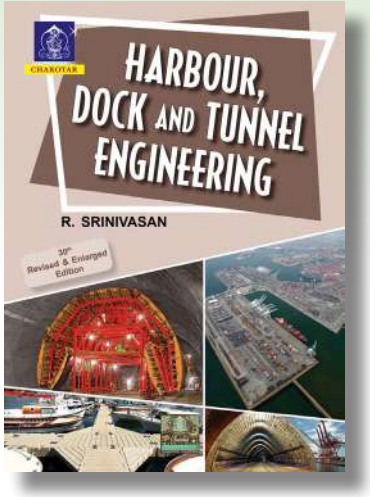

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HARBOUR DOCK AND TUNNEL ENGINEERING

NEW

REVISED & ENLARGED

By
R. Srinivasan

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ABOUT THE BOOK

In the subject of Transportation Engineering, study of Harbour, Dock and Tunnel Engineering is essential. This well-known text-book now in its twenty-ninth thoroughly Revised and Enlarged edition, concisely formulates the basic principles of the subject matter in simple, lucid and easy language presented in two sections.

Section I – Harbour and Dock Engineering

is well-divided in twelve chapters. It discusses the topics on Harbour and Ports; Natural Phenomena: Tides, Wind and Waves; Protection facilities: Mound Breakwater, Wall Type Breakwater, Special Type Breakwater such as Floating Type, Hydraulic Type and Pneumatic Type Breakwater; Planning and Layout of Ports; Various other Facilities such as Docking Facilities (Wet Docks, Basins and Lock Gates); Repairing Facilities (Dry Docks, Slipways and Floating Docks); Approach Facilities (Entrance and Lock Gates); Loading Unloading Facilities (Quay Walls, Wharves, Piers, Dolphins, Jetties and Fenders); Storing Facilities (Aprons, Transit Sheds and Warehouses); Dredging Facilities and Guiding Facilities.

Section II – Tunnel Engineering

is also well-divided in twelve chapters. It discusses the topics on General Aspects of Tunnelling; Stages in Tunnel Construction; Shafts and Portals; Soil Classification and Various Tunnelling Methods for Soft Soils and Other Methods of Tunnelling; Tunnelling in Water Bearing Soils; Tunnelling in Rock; The New Austrian Tunnelling Method (NATM) — Sequential Excavation Method; Tunnel Lining; Drainage of Tunnels; Lighting, Ventilation and Dust Control in Tunnels; and Safety in Tunnel Construction.

The Salient Features of this book are

243 Self explanatory neatly drawn sketches, photographs and more than 321 examination questions at the end of each chapter.

The book in the present form will prove to be extremely useful to the students preparing for the Degree examinations in Civil Engineering of all the Indian Universities, Diploma examinations conducted by various Boards of Technical Education, Certificate Courses as well as for the A.M.I.E., U.P.S.C., G.A.T.E., I.E.S., and other similar competitive and professional examinations. It should also be an immense use to practicing Civil Engineers.

CONTENT

SECTION I : HARBOUR AND DOCK ENGINEERING

1 : HARBOURS AND PORTS
2 : NATURAL PHENOMENA: TIDES, WIND AND WAVES
3 : PROTECTION FACILITIES: MOUND BREAKWATER
4 : PROTECTION FACILITIES: WALL TYPE AND SPECIAL BREAKWATERS
5 : PLANNING AND LAYOUT OF PORTS
6 : DOCKING FACILITIES (WET DOCKS, BASINS, LOCK GATES)
7 : REPAIRING FACILITIES (DRY DOCKS, SLIPWAYS, FLOATING DOCKS)
8 : APPROACH FACILITIES (ENTRANCE AND LOCK GATES)
9 : LOADING UNLOADING FACILITIES (QUAY WALLS, WHARVES, PIERS, DOLPHINS, JETTIES, FENDERS)
10 : STORING FACILITIES (APRONS, TRANSIT SHEDS AND WAREHOUSES)
11 : DREDGING AND DREDGING EQUIPMENT
12 : GUIDING FACILITIES

SECTION II : TUNNEL ENGINEERING

13 : GENERAL ASPECTS OF TUNNELLING
14 : STAGES IN TUNNEL CONSTRUCTION
15 : SHAFTS AND PORTALS
16 : SOIL CLASSIFICATION AND TUNNELLING METHODS FOR SOFT SOILS
17 : OTHER METHODS OF TUNNELLING
18 : TUNNELLING IN WATER BEARING SOILS
19 : TUNNELLING IN ROCK
20 : THE NEW AUSTRIAN TUNNELLING METHOD (NATM) [SEQUENTIAL EXCAVATION METHOD]
21 : TUNNEL LINING
22 : DRAINAGE OF TUNNELS
23 : LIGHTING, VENTILATION AND DUST CONTROL IN TUNNELS
24 : SAFETY IN TUNNEL CONSTRUCTION

INDEX: HARBOUR AND DOCK ENGINEERING
INDEX: TUNNEL ENGINEERING

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HARBOUR, DOCK AND TUNNEL ENGINEERING
DETAILED CONTENTS

SECTION I HARBOUR AND DOCK ENGINEERING

CHAPTER 1 HARBOURS AND PORTS

1-1. Introduction
1-2. Water transportation
1-3. Evolution of water transport
HARBOURS
1-4. Classification of harbours
1-4-1, Classification of harbour depending upon the protection needed
1-4-2. Classification of harbour depending upon the utility
1-4-3. Classification of harbour based upon the location
1-5. Accessibility and size of harbours
1-5-1. Site selection
1-5-2. Shape of the harbour
1-5-3. Harbour depth
1-5-4. Marine surveys
1-5-5. Harbour planning
1-5-6. Features of a port
1-5-7. Defects in harbours
1-5-8. Requirements or characteristics of a good harbour
1-5-9. Ship characteristics
1-5-10. Common terms for ship characteristics
PORTS
1-6. Ports
1-7. Indian Ports
1-8. Requirements of a good port
1-9. Facilities at a major port
1-10. Port design
1-11. Types of ships
Questions 1

CHAPTER 2 NATURAL PHENOMENA: TIDES, WIND AND WAVES

2-1. General
2-2. Littoral drift
2-3. Sea water waves
2-4. Tide generation
Tidal waves and tidal theory
2-5. Types of Lunar tides
2-6. Solar tides
2-7. Tides due to moon and sun
2-8. Total number of tides
2-9. Major tides
Spring tides and neap tides
2-10. Water levels of sea during tides and tidal ranges
2-11. Uses of tides
2-12. Effect of tides
2-13. Age or establishment of tide
2-14. Tide prediction
2-15. Wind
2-16. Dynamic effect of wave action
2-17. Modification of sea waves
2-18. Air compression
2-19. Water hammer
2-20. Harbour maintenance / Shore protection works
Questions 2

CHAPTER 3 PROTECTION FACILITIES

mound breakwater
3-1. General
3-2. Classification of breakwaters
3-3. Heap or mound breakwater
3-4. Selection of type of breakwater
Functions of components of mound breakwater
3-5. Details of energy dissipation in mound breakwater
3-6. Characteristics of mound breakwater
3-7. Rubble mound Breakwater
3-8. Concrete block mound breakwater
3-9. Rubble mound breakwater supplemented by concrete blocks
3-10. Rubble mound breakwater supplemented by patented stones
3-11. Mound with superstructure or Composite breakwater
Mound construction with top part made solid
3-12. Mound construction
Questions 3

CHAPTER 4 PROTECTION FACILITIES: WALL TYPE AND SPECIAL BREAKWATERS

4-1. Wall type breakwater
4-2. Types of wall breakwater
4-3. Causes of failure of wall breakwater
4-4. Forces on upright wall breakwater
4-5. Essentials for wall type breakwater
4-6. Typical cross-section of wall breakwater
Alternate cross-section of wall breakwater
4-7. Advantages and disadvantages of wall breakwater
4-8. Method of construction – staging system
4-9. Bonds for wall breakwater
4-10. Wall type breakwater of larger units
4-11. Breakwater height
4-12. Comparison of mound type and wall type breakwaters
4-13. Breakwater failures
4-14. Special breakwaters
Questions 4

CHAPTER 5 PLANNING AND LAYOUT OF PORTS

5-1. General
5-2. Facilities at a port
5-3. Layout of ports
Questions 5

CHAPTER 6 DOCKING FACILITIES (WET DOCKS, BASINS, LOCK GATES)

6-1. General
6-2. Classification of Docks
6-3. Classification of wet docks
6-4. Advantages and disadvantages of tidal wet docks
6-5. Advantages and disadvantages of enclosed wet docks or impounded basins
6-6. River ports
6-7. Form and arrangements of basins and docks
6-8. Design and construction of basin or dock walls
6-9. Other aspects of construction details of basin or dock walls
6-10. Dock entrances
6-11. Sizes of dock entrances
Questions 6

CHAPTER 7 REPAIRING FACILITIES (DRY DOCKS, SLIPWAYS, FLOATING DOCKS)

7-1. General
7-2. Classification of repairing facilities
7-3. Graving dry dock
7-4. Facilities to be provided at a graving dry dock
7-5. Method of dry docking
7-6. Size of graving dock
7-7. Forces acting on a graving dock
7-8. Conditions for design of graving dock
7-9. Constructing graving dock
7-10. Floor design of graving dry dock
7-11. Advantages and disadvantages of graving dry dock
7-12. Marine railway dry dock

7-13. Slipways
7-14. Lift dry dock
7-15. Floating type dry dock
7-15-1. Types of floating docks
7-15-2. Design considerations for floating docks
7-15-3. Advantages and disadvantages of floating dock
Questions 7

CHAPTER 8 APPROACH FACILITIES (ENTRANCE AND LOCK GATES)

8-1. General
8-2. Direction of an entrance of harbour
8-3. Direction of entrance for river harbours
8-4. Dimensions of entrances
8-5. Types of entrances
8-6. Entrance locks
8-7. Lock foundations
8-8. Dimensions of entrances and locks
8-9. Construction of lock gates
8-10. Types of lock gates
8-11. Forces on the gates
8-12. Shape of gates
8-13. Support for dock gates
8-14. Working of gates
Questions 8

CHAPTER 9 LOADING-UNLOADING FACILITIES (WALLS, WHARVES, PIERS, DOLPHINS, JETTIES, FENDERS)

9-1. General
9-2. Design of quay walls
9-3. Types of quay walls
9-4. Other details of quay walls
9-5. Wharves
9-6. Piers
9-7. Types of piers
9-8. Additional points for Piers
9-9. Pierheads
9-10. Dolphins
9-11. Jetties
9-12. Differences between wharf and jetty
9-13. Fenders
9-13-1. Basic features of marine fenders
9-13-2. Classification of fenders
9-13-3. Type of fenders
9-14. Slip
9-15. Moles
Questions 9

CHAPTER 10 STORING FACILITIES (APRONS, TRANSIT SHEDS, AND WAREHOUSES)

10-1. General
10-2. Aprons
10-3. Transit sheds
10-4. Design of transit shed
10-5. Warehouses location of warehouse
10-6. Cold storages
10-7. Guard houses
Questions 10

CHAPTER 11 DREDGING AND DREDGING EQUIPMENT

11-1. Introduction
11-2. Purpose of dredging
11-3. Application of dredging required areas
11-4. Classification of dredging
11-4-1. Primary dredging or Capital dredging
11-4-2. Secondary dredging or Maintenance dredging
11-5. Objectives of dredging
11-6. Types of dredged materials
11-7. Disposal of the dredged material
11-8. Types of dredging equipment
11-8-1. Mechanical dredgers
11-8-2. Hydraulic dredgers
11-8-3. Specialised types of dredgers
11-9. Choice of dredger
11-10. Execution of dredging work
Questions 11

CHAPTER 12 GUIDING FACILITIES

12-1. Necessity for guiding facilities
12-2. Fixed and floating light stations
12-3. Lighthouse
12-4. Signals
12-5. Light signals
12-6. Fog signals
12-7. Audible signals
12-8. Moorings
12-9. Mooring accessories
12-10. Off-shore moorings
Questions 12

SECTION II TUNNEL ENGINEERING

CHAPTER 13 GENERAL ASPECTS OF TUNNELLING

13-1. General
13-2. Categories of obstacles
13-3. Definitions
13-4. Comparison of bypassing alternatives tunnel, open cut, bridge and surface road
13-5. Advantages and disadvantages of tunnels and open cuts
13-6. History of tunnels constructed
13-7. Developments in tunnelling methods
13-8. Important years in tunnel construction
13-9. Economics of tunnelling
13-10. Alignment of a tunnel
13-11. Classification of tunnels
13-12. Tunnel approaches
13-13. Shapes of tunnels
13-14. Size of tunnels
13-15. Problems in tunnelling
Questions 13

CHAPTER 14 STAGES IN TUNNEL CONSTRUCTION

14-1. Investigations at tunnel site
14-2. Setting out of tunnel
14-3. Methods of getting extra faces to work upon tunnel
14-4. Excavation
14-5. Blasting
14-6. Temporary supports
Temporary supports for steel
14-7. Permanent supports
14-8. Ventilation at the time of construction
14-9. Muck removal
14-10. Supplementary operations
14-11. Miscellaneous
Questions 14

CHAPTER 15 SHAFTS AND PORTALS

15-1. General
15-2. Advantages of shafts
15-3. Size of shafts
15-4. Location of shafts
15-5. Classification of shafts

15-6. Construction of shafts in rock
15-7. Construction of shaft in soft ground
15-8. Design of shaft supports
15-9. Precautions for shaft sinking work in soft soil
15-10. Protection round the shaft opening
15-11. Portals
15-12. Twin tunnels
Questions 15

CHAPTER 16 SOIL CLASSIFICATION AND TUNNELLING METHODS FOR SOFT SOILS

16-1. Soil classification
16-2. Choice of tunnelling methods in soft soils
16-3. Methods of tunnelling (soft soils)
16-4. Forepoling method
16-5. Needle beam method
16-6. Army method or case method
16-7. American method
16-8. English method
16-9. Belgian method
16-10. German method
16-11. Austrian method
16-12. Timbering in soft soil tunnelling
Questions 16

CHAPTER 17 OTHER METHODS OF TUNNELLING (LINER PLATE METHOD)

17-1. Liner plate method
CUT AND COVER TUNNEL CONSTRUCTION METHOD
17-2. Cut and cover tunnel construction method
17-3. Features of cut and cover tunnel construction methods
17-4. Types of cut and cover tunnel construction methods
SHIELD AND TUNNEL BORING MACHINE METHOD
17-5. Tunnelling with shield
17-5-1. Parts of shield
17-5-2. Terms commonly used with shield
17-5-3. Primary lining in shield
17-5-4. General steps of tunnelling with shield
17-5-5. Shield tunnelling in different types of soils
17-5-6. Common equipment with shield
17-5-7. Sequence of tunnelling using the shield
17-5-8. Mechanized shields
17-6. Mechanised tunnel boring machine
17-7. Factors for selection of TBM
17-8. Sequence of mechanised TBM
17-9. Types of mechanised TBM
17-10. Open type TBM for hard rock
17-10-1. Open type TBM without shield
17-10-2. Enlargement tunnel boring machines (ETBM)
17-11. Double shield (open face) tunnel boring machine (DSTBM)
17-12. Closed type single shield TBM for soft soil
17-12-1. Slurry pressure tunnel boring machine (SPTBM)
17-12-2. Earth pressure balance tunnel boring machine (EPBM)
17-13. Multi-mode tunnel boring machine (MMTBM)
17-14. Difference between open type TBM and closed type shield TBM
17-15. Difference between single shield TBM and double shield TBM
Questions 17

CHAPTER 18 TUNNELLING IN WATER BEARING SOIL

18-1. General
18-2. Well points system Procedure to bring down ground water table
18-3. Equipment with Plenum process of tunnelling or compressed air method
18-4. Use of pipes and conduits in water bearing soils
18-5. Compressors, generators and pumps
18-6. Methods of tunnelling in water bearing soils
Questions 18

CHAPTER 19 TUNNELLING IN ROCK

19-1. General
19-2. Sequence of operations for tunnelling in rock
19-3. Faces of operation for tunnelling in rock
19-4. Methods of tunnelling in rock
19-5. Mucking
19-6. Mucking in steep grade tunnelling
19-7. Hauling
19-8. Other aspects
19-9. Drill-bits
19-10. Nipper cars
17-11. Explosives
17-12. Safety precautions in rock tunnelling
Questions 19

CHAPTER 20 NEW AUSTRIAN TUNNELLING METHOD (NATM) [SEQUENTIAL EXCAVATION METHOD]

20-1. Introduction to NATM
20-2. Objectives of NATM
20-3. Features Based on NATM
20-4. Components and sequence of execution in NATM
20-5. Conclusions
Questions 20

CHAPTER 21 TUNNEL LINING

21-1. Necessity of lining
21-2. Objects of tunnel lining
21-3. Materials for lining
21-4. Design of thickness of lining
21-5. The sequence of lining a tunnel
Questions 21

CHAPTER 22 DRAINAGE OF TUNNELS

22-1. General
22-2. Pre-drainage
22-3. Dewatering
22-4. Permanent drainage
Questions 22

CHAPTER 23 LIGHTING, VENTILATION AND DUST CONTROL IN TUNNELS

23-1. Tunnel Lighting
23-1-1. Spacing of lights
23-1-2. Types of tunnel lights
23-2. Ventilation in tunnels
23-2-1. Objects of tunnel ventilation
23-2-2. Requirements of tunnel ventilation
23-2-3. Volume of air required
23-2-4. Methods of ventilation
23-2-5. Equipment required for tunnel ventilation
23-2-6. Permanent ventilation and noise pollution
23-3. Dust control and methods
Questions 23

CHAPTER 24 SAFETY IN TUNNEL CONSTRUCTION

24-1. General
24-2. Safety precautions in tunnelling
24-3. Health protection in tunnel construction
Questions 24

INDEX HARBOUR AND DOCK ENGINEERING

INDEX TUNNEL ENGINEERING